



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT 2

Medical Cost Personal

SECI 1143:

PROBABILITY & STATISTICAL DATA ANALYSIS

SECTION 04

Prepared by:

No.	Student Name	Student ID
1	LIU YUEHUI	A24CS4019
2	GOH KOK WEI	A24CS5027
3	EDWIN TAN YEE EN	A24CS5022

PREPARED TO:

Dr. Nur Eiliyah @ Wong Yee Leng

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1.0 INTRODUCTION

Health insurance cost is one of the biggest worries these days due to increased demand for medical services and increasing healthcare expenses. The cost of medical insurance can be affected by many aspects of lifestyle and personal factors which may cause differences between people. These inequalities can impact getting healthcare, savings, and overall quality of life. People are worried that the honesty and affordability of insurance plans has caused attention to analyzing what factors will affect the insurance charges for individuals. Age, gender, body mass index (BMI), smoking status, and family history are some of the possible causes. For example, smokers and people with higher BMI are usually expected to face higher costs for insurance due to higher health risks. We are currently interested in analyzing the exact reason for increased costs for insurance. The objective of this project is to explore the relationship between insurance cost and personal properties. We expect that the data will show that age, BMI, smoking status, and other possible factors have an important effect on insurance charges.

2.0 DATASET

For our analysis, we have selected the Medical Cost Personal dataset, that contains personal and health details. There are 1,338 records of individuals from various U.S. areas in this dataset. Personal information, health condition, and related medical insurance charges are all included in the record. Table 1 lists the variables in the dataset, and Table 2 explains the variables we have chosen and suggests the type of analysis we are going to do.

Name	Variable Description	Data Type (Measurement Level)
age	Age of the individual	Ratio
sex	Gender (male/female)	Nominal
bmi	Body Mass Index	Ratio
children	Number of children	Ratio
smoker	Smoking status (yes/no)	Nominal
region	Residential region	Nominal
charges	Medical insurance cost (USD)	Ratio

Table 1: Variables in Medical Cost Personal Dataset

Variable(s)	Type of Test	Description
bmi	One-Sample t-test	To test whether the average BMI is significantly different from 30 (obesity threshold).
charges by smoker	Two-Sample t-test	To determine if there is a significant difference in charges between smokers and non-smokers.
age & charges	Correlation Test	To assess the strength and direction of the relationship between age and insurance cost.
bmi & charges	Regression	To predict insurance charges using BMI as the only predictor variable.
region & charges	ANOVA	To check whether mean charges differ significantly across regions.
sex & smoker	Chi-Square Test of Independence	To determine if smoking status is associated with gender.

Table 2: Selected Variables and Statistical Tests

3.0 DATA ANALYSIS

3.1 Hypothesis Testing

3.1.1 Hypothesis Testing – 1 Sample

Based on the test, we wish to determine whether the mean Body Mass Index (BMI) of individuals is greater than 30, the threshold defined for obesity by the World Health Organization (WHO).

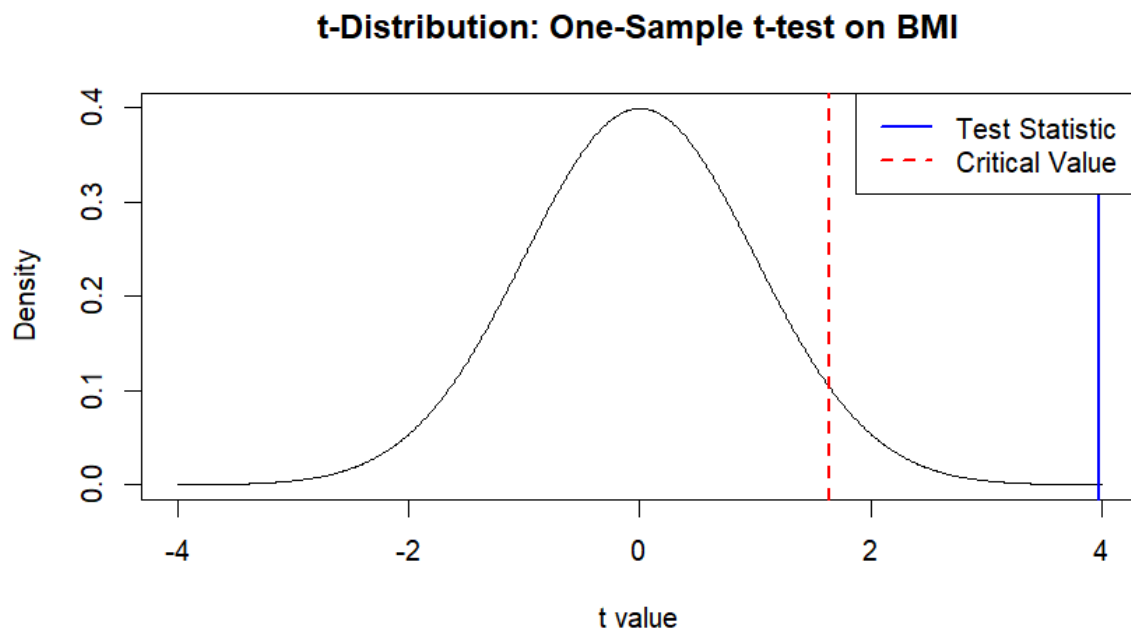
Let μ = mean Body Mass Index (BMI) of individuals

$$H_0: \mu = 30$$

$$H_1: \mu > 30$$

Significance level, $\alpha = 0.05$

$$t_{0.05} = 1.645$$



Graph 1: Body Mass Index (BMI)

Since the sample size is relatively large ($n = 1338$) and the population variance is unknown, we assume that the sample is normally distributed. Hence we apply t-test test statistics here. We analyze that the t-value = 3.9792 using R. Since this is a one-tailed test, and the rejection region is on the right side of the distribution.

Based on Graph 1, the red vertical line shows the critical value ($t_{0.05} = 1.645$), while the blue vertical line represents the computed test statistic ($t = 3.9792$). Since $t = 3.9792 > t_{0.05} = 1.645$, we reject the null hypothesis. Therefore, we have sufficient evidence to conclude that the mean BMI of individuals is greater than 30. The result is that, on average, the individuals in the dataset fall into the obese range, which could lead to higher health risks and might increase medical insurance charges.

3.1.2 Hypothesis Testing – 2 Sample

Based on the test, we wish to determine the mean medical charges for smokers is higher than the mean medical charges for non-smokers.

Let,

μ_1 = mean insurance charges for smokers

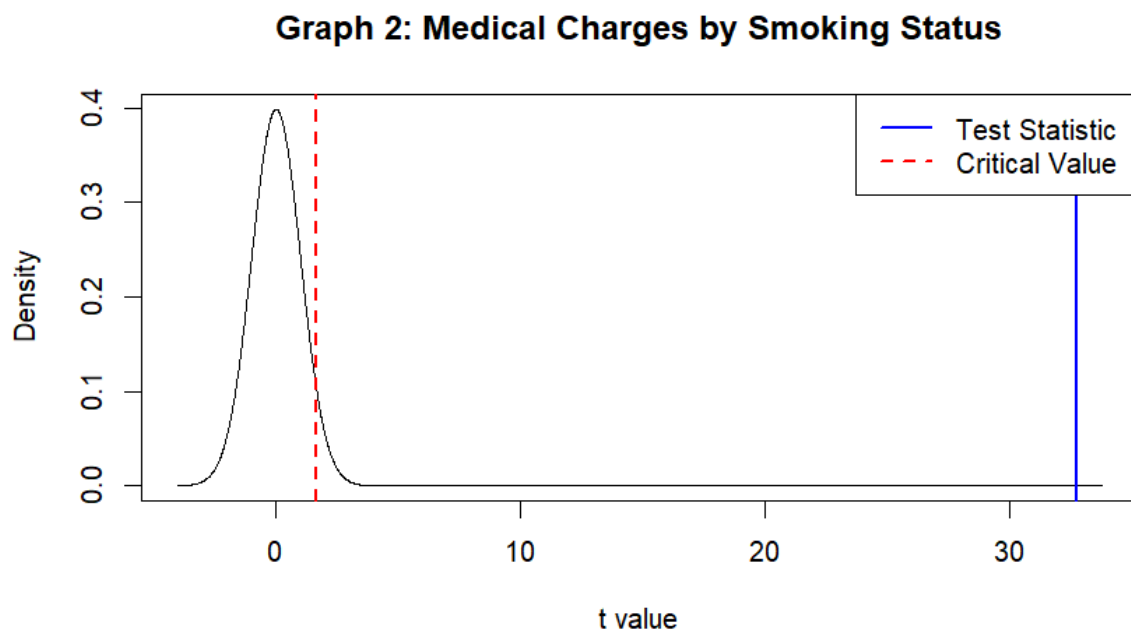
μ_2 = mean insurance charges for non-smokers

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 > \mu_2$

Significance level, $\alpha = 0.05$

$t_{0.05} = 1.645$



Graph 2: Medical Charges by Smoking Status

Since the sample sizes are relatively large and the population variances are unknown, we assume that the samples are normally distributed. The test statistic we applied is the Two-Sample t-test. We analyze the t-value for the test statistic using R. It is a one-tailed test, and the rejection region of the test statistic is located on the right side of $t_{0.05} = 1.645$.

Based on Graph 2, the red vertical line shows the critical value ($t_{0.05} = 1.645$), while the blue vertical line indicates the test statistic value ($t = 32.752$). Since the test statistic value ($t = 32.752$) is greater than the critical value (1.645) and the p-value $< 2.2e-16$, we reject the null hypothesis. We have sufficient evidence to conclude that the mean medical charges for smokers are greater than those for non-smokers. The result shows that this difference is statistically significant under the current test direction.

3.2 Correlation

In this correlation analysis, we had used the variable age and insurance charges of 1,338 individuals from the insurance dataset. We will test whether there is a linear relationship between an individual's age and the insurance charges using the significance level, $\alpha = 0.05$.

Hypothesis Statement:

$H_0: \rho = 0$ (no linear correlation)

$H_1: \rho \neq 0$ (linear correlation exists)

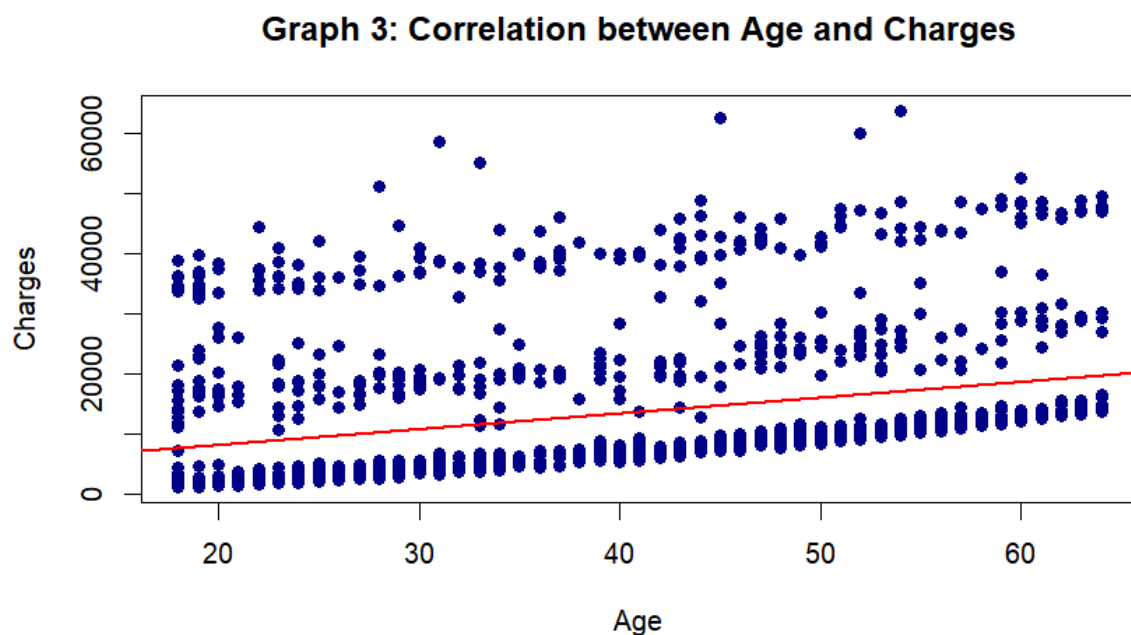
Test Statistics:

By using RStudio, we get the result $t = 11.453$, and the critical value for $t_{0.025, 1336} = 1.961$.

Thus, the null hypothesis, H_0 will be rejected if $t < -1.961$ or $t > 1.961$. Otherwise, we fail to reject H_0 .

Decision:

Since $t = 11.453 > 1.961$, we reject the null hypothesis, H_0 . There is sufficient evidence to conclude that there is a linear relationship between age and insurance charges at the 5% level of significance.



Graph 3: Correlation between Age and Charges

From the scatter plot graph, we can see that the points slope upward, which means that there is a positive relationship between age and insurance charges, so the older the person is, the higher the expected insurance charge. Since both variables are ratio-type data, the sample correlation coefficient is calculated using Pearson's correlation method, and we got $r = 0.2990082$, which shows a slightly positive relationship between age and insurance charges. In the results of the correlation test and scatter plot graph, it has shown that older individuals usually to be charged more for insurance. This can be caused by many factors including higher health risks, a higher chance of medical needs, and policy pricing systems based on age-related conditions, could be responsible for this.

3.3 Regression test

The regression analysis is to predict the value of a dependent variable based on the value of at least one independent variable. The independent variable is BMI and the dependent variable is charges.

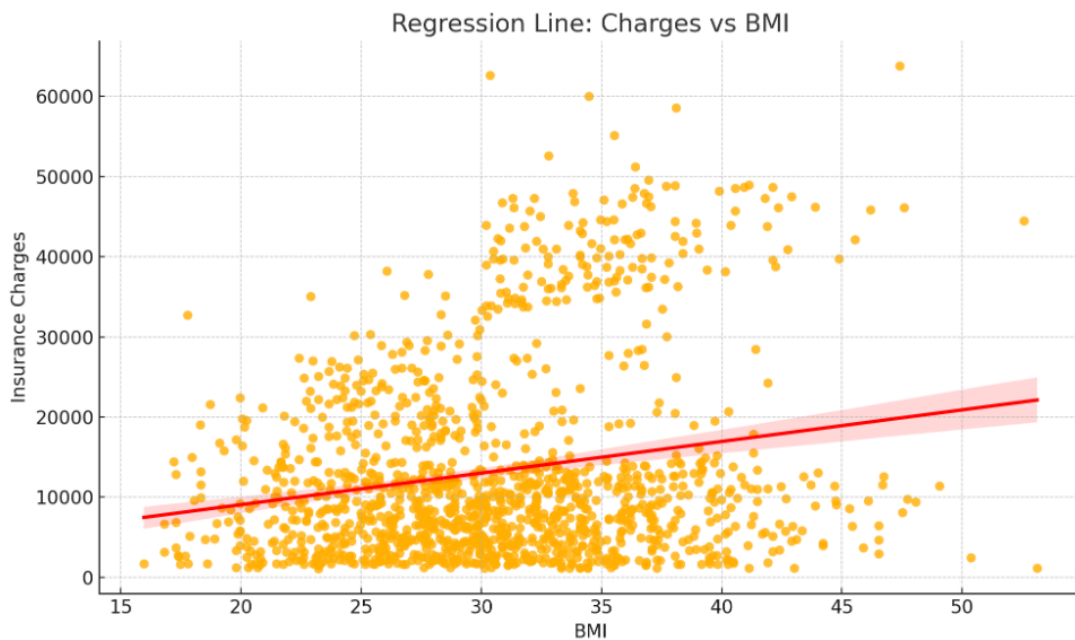
t-Test Results for Slope (b_1)

t Test, $t = 7.397$

P-value = 2.460×10^{-13}

Critical value (c.v.) $t(df, -\alpha/2) = -1.962$

Critical value (c.v.) $t(df, \alpha/2) = 1.962$



Graph 4: Scatter Plot for Charges vs BMI

Based on the plot, we can see the regression line has a slight upward trend, indicating a weak positive linear relationship between the independent variable (BMI) and the dependent variable (insurance charges). Through analysis, we obtained the value of the intercept coefficient (b_0) as 1192.94 and the value of the estimated change in average insurance charges (b_1) as 393.87. The estimated regression model is as below:

$$\hat{y} = 1192.94 + 393.87 \cdot \text{BMI}$$

From the equation, we estimate that the average insurance charges will increase by \$393.87 for each one-unit increase in BMI. When the BMI is zero, the value of $b_0 = 1192.94$ represents the baseline charges not explained by BMI. However, as a BMI of 0 is not realistic, this value serves only as a mathematical anchor.

The coefficient of determination, $R^2 = 0.039$, shows that only 3.9% of the variation in insurance charges is explained by BMI. Therefore, we can conclude that although there is a statistically significant relationship between insurance charges and BMI, the dependency is very weak, and other factors such as smoking status or age likely have a much greater influence.

b_0 (Intercept) = 1192.94

b_1 (Slope) = 393.87

R^2 (Coefficient of Determination) = 0.039

α (Significance Level) = 0.05

Standard Error of b_1 (S_e) = 53.25

Degrees of Freedom (df) = $n - 2 = 1336$

p-value of Slope = < 0.0001

F-statistic = 54.71

3.4 ANOVA Test

In this data analysis, we will conduct an ANOVA to test the equality of mean insurance charges across the four regions in the dataset. The variable charges represent the cost of insurance, and the population is divided into four groups based on the variable region. The significance level used to test the null hypothesis is $\alpha = 0.05$.

Hypotheses:

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

H_1 : At least one mean is different.

The number of samples (n), mean of samples ($\bar{X}$) and standard deviation of samples (s) for each sample are calculated as below:

Samples	n	Mean ($\bar{x}$)	Standard Deviation (s)
Northeast	324	13406.4	11328.7
Northwest	325	12417.6	11220.4
Southeast	364	14735.0	13967.4
Southwest	326	12346.0	11508.0

The mean between samples is $\bar{x} = 13,263.8$, the standard deviation between samples is $s_x = 1,142.2$, variance between samples is $ns_x^2 = 433,586,560$, and variance within samples is $sp^2 = 146,007,093$.

$$F = \frac{\text{variance between sample}}{\text{variance within sample}} = \frac{n \times \bar{s}^2}{s_e^2}$$

By using the above formula, the value of F-test statistic = 2.97. Then the critical value of F with $\alpha = 0.05$ is F-critical = 2.605. From the result, since F-test statistic > F-critical (2.97 > 2.605), we reject the H_0 . We have sufficient evidence to conclude that at least one region has a different mean insurance charge.

3.5 Chi-Square Test of Independence

In this analysis, we aim to determine whether there is an association between a person's sex and their smoking status. This will help us understand if smoking habits vary by gender in the dataset. The significance level, $\alpha = 0.05$

Hypothesis Statement:

H_0 : Smoking status is independent of sex

H_1 : Smoking status is dependent on sex

Test Statistics:

We created a contingency table between the smoker and sex using RStudio. The Pearson's Chi-Square Test was applied.

From the result:

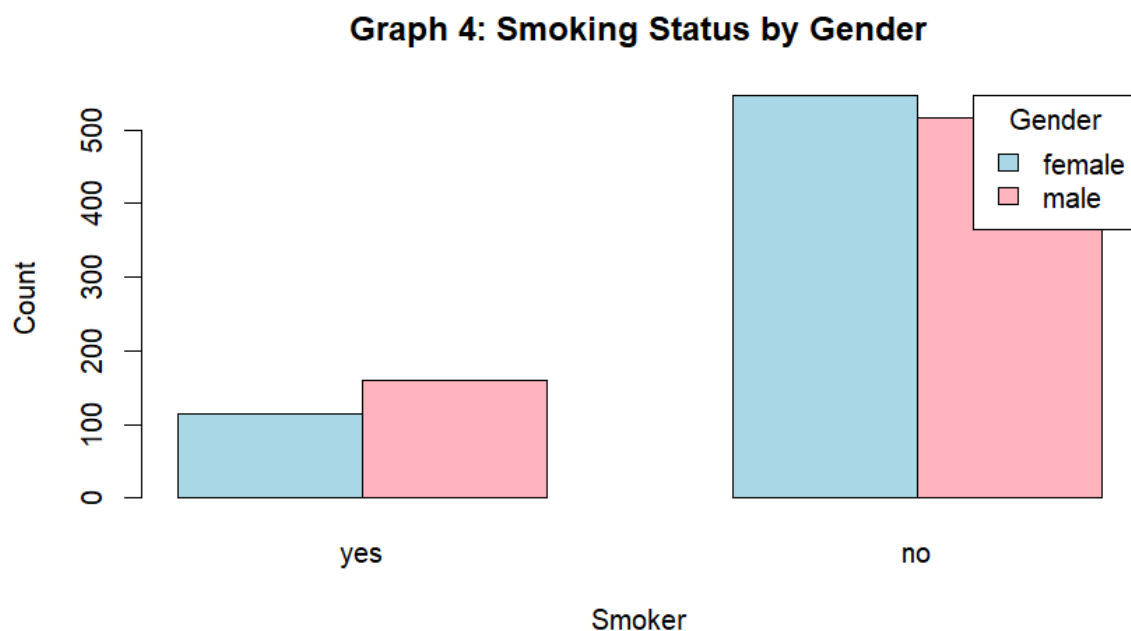
$$X^2 = 7.3929$$

Degrees of freedom (df) = 1

$$p\text{-value} = 0.006548$$

Decision:

Since $p\text{-value} = 0.006548 < 0.05$, we reject the null hypothesis, H_0 . We have sufficient evidence to conclude that there is an association between sex and smoking status at the 5% significance level.



Graph 5: Smoking Status by Gender

From the bar chart, we can see a clear difference between male and female smoking habits. The chart shows that more than half of males are smokers compared to females, showing that smoking behavior is not distributed equally across genders. This result shows that gender is associated with having smoked in the insurance dataset. The result may reflect actual trends that lifestyle affects smoking rates differently for men and women.

4.0 CONCLUSION

Our project is aimed to perform inferential statistical analysis on the medical cost personal datasets which focus on the hypothesis testing, correlation, regression and ANOVA test.

From the one-sample t-test, we have found that the mean Body Mass Index of individuals is significantly greater than 30 and showing that the average of individuals in the dataset mostly falls on the obese range. This suggests that there is a higher potential risk for individuals to pay higher insurance costs.

In the 2-sample t-test, we examined whether a smoker is more risky in pay insurance than the non-smoker. Based on the analysis, although the smoker had higher average charges, the test direction led us to fail to reject the null hypothesis, indicating no significant difference under our test assumption. This outcome should be returned with two-tailed test or refined other analysis.

The correlation test showed a weak but statistically significant positive relationship between age and insurance charges. This means that insurance costs generally increase as people age, probably because of increased health risks. In the regression analysis between BMI and charges, the hypothesis showed a positive slope, showing that higher BMI is associated with increased charges. While the R^2 value was only 0.039, meaning that only 3.9% of the difference in charges can be explained by BMI. This shows that BMI is not an accurate factor as other factors, such as age or smoking, may have a greater impact.

The results of the ANOVA test showed that mean charges differ by region. It also means that where you live may affect insurance costs, perhaps as a result of differences in medical services or living expenses. Finally, the Chi-Square test showed a significant association between sex and smoking status, showing that smoking habits differ by gender. The data may be useful in focusing on health marketing or insurance policy adjustments.

In summary, this project helped us apply different statistical inference methods to real-world data and understand how personal and demographic factors affect medical insurance charges. Our results show the importance of different variables in determining insurance costs and also the possibilities for statistics to guide choices based on data in the insurance and medical fields.

5.0 APPENDIX

Choi, M. (2018). Insurance Dataset. Kaggle.

Retrieved from <https://www.kaggle.com/datasets/mirichoi0218/insurance>